

Arduino-Based Automated Radar Targeting System

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ABSTRACT

This project presents the design and implementation of an Arduino-based automated radar targeting system capable of detecting objects within a fixed 180-degree field of view and actuating a servo-mounted projectile launcher. The system integrates ultrasonic distance measurement, IR-based presence detection, servo motor control, and real-time graphical visualization using the Processing environment. For demonstration purposes, a water gun was used as the projectile launcher; however, the system architecture allows adaptation to other low-power launchers such as Nerf-based mechanisms with suitable mechanical modifications. The system operates autonomously once powered, although projectile reloading is manual and not automated.

1. INTRODUCTION

Radar and targeting systems are widely used in surveillance, robotics, and defense-oriented applications. This project aims to simulate a simplified radar-guided targeting system using low-cost and easily available components. By combining sensor-based detection with servo-driven actuation and a graphical radar interface, the project demonstrates fundamental concepts of embedded systems, sensor fusion, real-time control, and visualization.

The project focuses on automation of detection and actuation while maintaining safety by limiting the system to low-power projectile mechanisms.

2. OBJECTIVES

The primary objectives of this project are:

- To design a 180-degree radar scanning system using a servo-mounted ultrasonic sensor
 - To detect objects using a combination of ultrasonic and IR proximity sensors
 - To visualize real-time detection data using a radar-style graphical interface
 - To automatically actuate a servo-controlled projectile launcher upon target detection
 - To demonstrate embedded system integration and real-time control
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3. COMPONENTS USED

3.1 Hardware Components

1. Arduino Uno R3
2. HC-SR04 Ultrasonic Sensor
3. IR Proximity Sensor (Active-Low Output)
4. Two SG90 Servo Motors
5. Servo-mounted projectile launcher (water gun for demonstration)
6. External 5–6V battery pack for servo motors
7. Breadboard and jumper wires

3.2 Software Components

Arduino IDE

Processing IDE

Built-in Arduino Servo Library

4. SYSTEM ARCHITECTURE

The system consists of a radar servo responsible for sweeping the ultrasonic sensor across a fixed angular range. The ultrasonic sensor measures distance, while the IR sensor confirms object presence. Sensor data is processed by the Arduino and transmitted to a computer via serial communication. The Processing application receives this data and renders a tactical radar-style user interface. A second servo motor acts as a trigger mechanism for the projectile launcher when detection conditions are met.

5. WORKING PRINCIPLE

1. The radar servo continuously sweeps between 10 degrees and 170 degrees
 2. At each angle, the ultrasonic sensor measures the distance to nearby objects
 3. The IR sensor validates the presence of an object
 4. The Arduino transmits angle, distance, and detection status via serial communication
 5. Processing visualizes the data in a radar-style graphical interface
 6. When an object is detected within the predefined range, the trigger servo is actuated automatically
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6. CIRCUIT DESCRIPTION

The ultrasonic sensor's TRIG and ECHO pins are connected to Arduino digital pins for pulse generation and echo measurement. The IR sensor output is connected to a digital input pin and operates in active-low mode. Two servo motors are controlled using PWM signals from the Arduino. Both servos are powered using an external battery to ensure sufficient current, with a common ground connection between the Arduino and the battery.

7. SOFTWARE IMPLEMENTATION

The Arduino program handles sensor data acquisition, servo control, and serial communication. A state-based sweep logic ensures smooth bidirectional scanning within a fixed 180-degree sector without pauses. The Processing program receives serial data and renders a radar interface featuring sweep animation, distance markings, target indicators, telemetry graphs, and warning overlays.

8. RESULTS AND OBSERVATIONS

The system successfully detected objects within a range of 50 cm and displayed their angular position accurately on the radar interface. The automated actuation of the projectile launcher functioned reliably when detection conditions were met. The radar visualization responded smoothly to object movement, demonstrating effective real-time performance.

9. APPLICATIONS

Automated targeting demonstrations
Robotics and mechatronics projects
Radar and surveillance simulations
Embedded systems education
Human-machine interface visualization

10. LIMITATIONS

Detection range is limited
Only single-target response is supported
Object identification is not possible
Projectile reloading is manual
Performance may be affected by ambient lighting for the IR sensor

11. FUTURE SCOPE

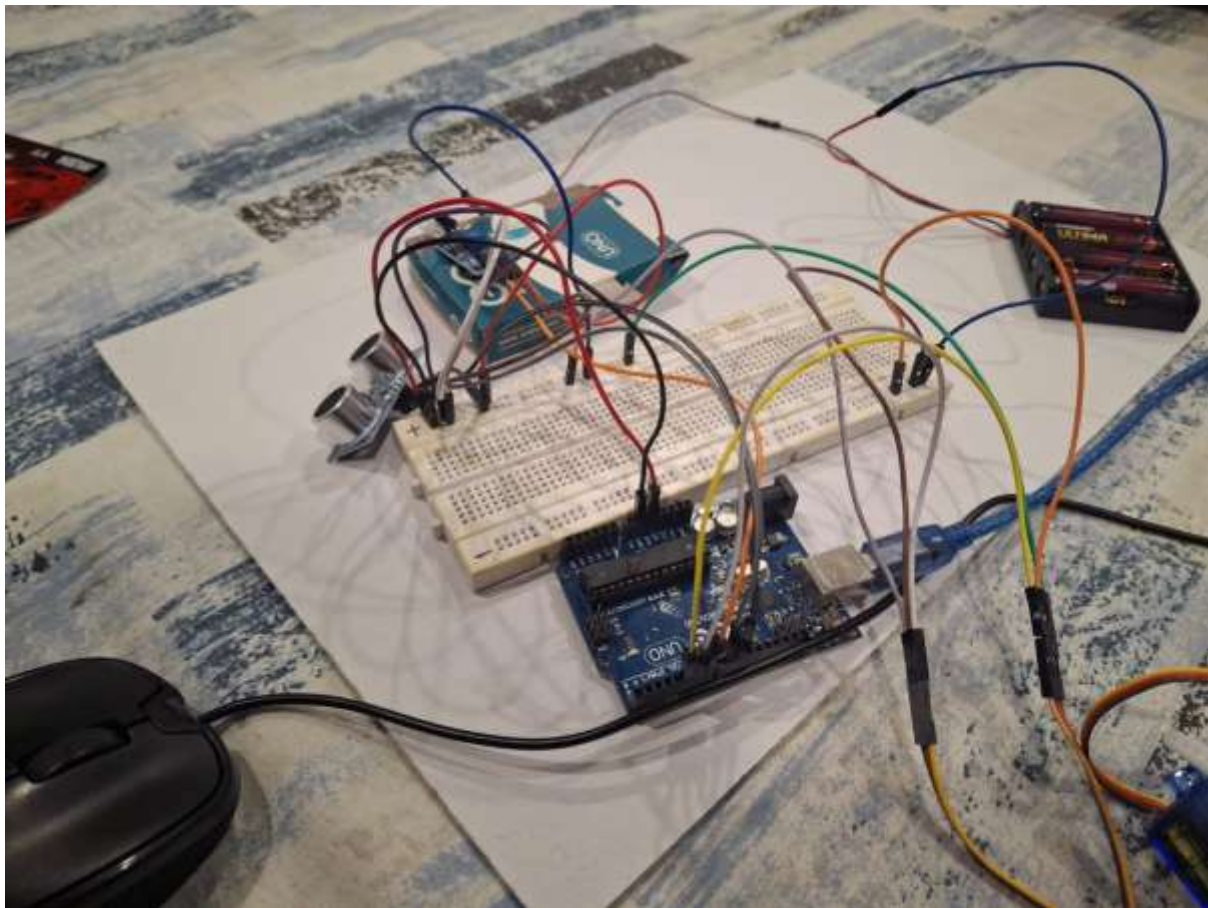
Implementation of lock-on target tracking
Multiple target identification and memory
Wireless monitoring and control
Improved launcher mechanisms
Advanced filtering and AI-based detection

12. CONCLUSION

The Arduino-based automated radar targeting system successfully demonstrates a functional combination of sensor fusion, servo control, and real-time visualization. The project highlights practical aspects of embedded system design while maintaining safety through the use of low-power projectile mechanisms. The modular design allows future expansion and improvement, making it suitable for educational and experimental applications.

REFERENCES

Arduino Official Documentation
HC-SR04 Ultrasonic Sensor Datasheet
Processing IDE Documentation
Servo Motor Control Principles



[Video of components and GUI working as intended.](#)

